

How Much Does It Cost to Tile a Pool?



The Garcia family is building a rectangular pool and wants to lay 1 ft x 1 ft tiles around the perimeter of the pool. They want the pool to be 8 feet wide, but they haven't decided yet how long it should be. They would like the pool to be at least 16 feet long, but given the size of their backyard, the maximum length is 22 feet.

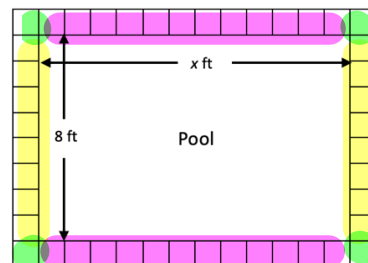
- How many tiles are needed if the length of the pool is 16 feet? Use colors to demonstrate your strategy for counting the tiles.

$$8(2) + 16(2) + 4 = 52 \text{ tiles}$$

Many methods possible:

$$10(2) + 16(2)$$

$$8(2) + 18(2)$$



- a. Use your strategy to determine how many tiles are needed if the length of the pool is x feet.

$$8(2) + 2x + 4 = 2x + 20 = n(x) \quad \text{input: length}$$

output: # of tiles

- b. What is the minimum number of tiles that would be needed? What is the maximum? How do you know?

Minimum is 52

$$\text{Max is } 2(22) + 20 = 64$$

Since x varies from 16 to 22, the # of tiles varies from 52 to 64

- Home Depot sells 1 ft by 1 ft tiles for \$5.75 per tile and charges a delivery fee of \$9.99. Write a rule for the total cost of having n tiles delivered.

$$9.99 + 5.75n = C(n) \quad \text{input: \# of tiles}$$

output: cost

- Fill in the table to help the Garcia family compare the cost of various size pools. Ignore sales tax. Show your work.

Side length	Number of Tiles	Cost
x	$n(x)$	$C(n(x))$
16 feet	$2(16) + 20 = 52$	$5.75(52) + 9.99 = \$308.99$
18 feet	$2(18) + 20 = 56$	$5.75(56) + 9.99 = \$331.99$
22 feet	$2(22) + 20 = 64$	$5.75(64) + 9.99 = \$377.99$

- Can you find a general rule for the cost of tiling a pool that's 8 feet wide by x feet long?

$$5.75(2x + 20) + 9.99 = C(n(x))$$

Input: length of pool
output: cost

Lesson 1.8 – Composition of Functions

QuickNotes

Composite functions are made up of 2 or more functions where the output of one function becomes the input of the next function.

$$y = f(g(x))$$

outside function inside function

"f of g of x"

The domain of $f(g(x))$ is the subset of g 's domain that will produce values in the domain of f .

Order matters! $f(g(x)) \neq g(f(x))$

Check Your Understanding

- The function $H(d)$ represents the amount of homework problems assigned, d days into the school year. The function $T(p)$ represents the amount of time it takes, in minutes, to complete p problems. Interpret the meaning of the statement $T(H(40)) = 128$ in context.

40 days into the school year, students can expect 128 minutes of homework.

- Find $f(g(x))$ and $g(f(x))$ for the pair of functions given and state the domain of each.

$$f(x) = 1/x, \quad g(x) = \sqrt{x-9}$$

$$\text{D of } g: x \geq 9$$

$$\text{D of } f: x \neq 0$$

$$\text{D of } f(g(x)): x > 9$$

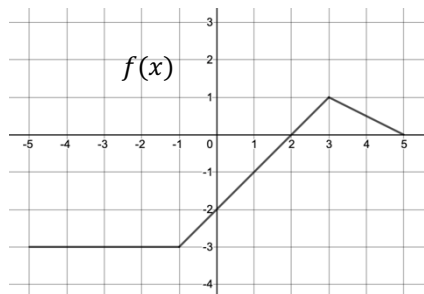
$$g(f(x)) = \sqrt{\frac{1}{x} - 9}$$

$$\frac{1}{x} \geq 9, \text{ and } x \neq 0$$

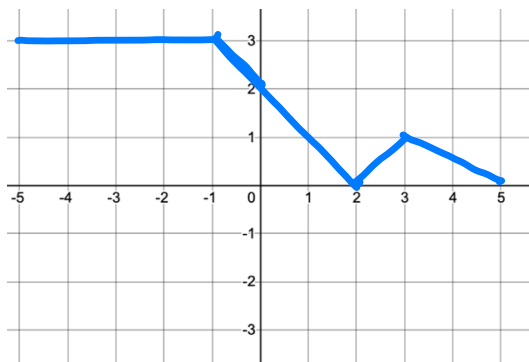
$$\text{Domain: } 0 < x \leq \frac{1}{9}$$

$$f(g(x)) = \frac{1}{\sqrt{x-9}}$$

- The graph of a function $f(x)$ defined for $-5 \leq x \leq 5$ is shown below. The function g is given by $g(x) = |x|$.



- Sketch the graph of $y = g(f(x))$.



- Sketch the graph of $y = f(g(x))$.

